

ESP32 Smart Turntable Wiring & Connection Guide

This document details the hardware connections and pin mappings for the ESP32 Smart Turntable system, integrating the SSD1306 OLED Display, MFRC522 RFID Module, HW-484 Hall Effect Sensor, and ULN2003 Stepper Motor Driver.

1. Master Connection Table

The following table provides a complete mapping of all peripheral connections to the ESP32 development board.

Peripheral Module	Module Pin	ESP32 Pin	Operating Voltage	Interface Type	Notes / Function
SSD1306 OLED	SDA	GPIO 21	3.3 V / 5 V	I ² C Data	Display Data Line (400 kHz Fast Mode)
	SCL	GPIO 22	3.3 V / 5 V	I ² C Clock	Display Clock Line
	VCC	3.3V	3.3 V	Power	Power supply input
	GND	GND	0 V	Ground	Common system ground
MFRC522 RFID	SDA / SS	GPIO 5	3.3 V ONLY	SPI Chip Select	Slave Select
	SCK	GPIO 18	3.3 V	SPI Clock	Bus Clock
	MISO	GPIO 19	3.3 V	SPI Data Out	Master In Slave Out
	MOSI	GPIO 23	3.3 V	SPI Data In	Master Out Slave In
	RST	GPIO 4	3.3 V	Control	Module Reset Pin
	3.3V	3.3V	3.3 V ONLY	Power	Do NOT connect to 5 V
	GND	GND	0 V	Ground	Common system ground
HW-484 Hall Sensor	DO	GPIO 27	3.3 V / 5 V	Interrupt (GPIO)	Digital Output (Tonearm position)
	VCC	3.3V	3.3 V	Power	Sensor power input
	GND	GND	0 V	Ground	Common system ground
ULN2003 Driver	IN1	GPIO 26	5 V Logic	Digital Out	Stepper Phase A
	IN2	GPIO 32	5 V Logic	Digital Out	Stepper Phase B

IN3	GPIO 33	5 V Logic	Digital Out	Stepper Phase C
IN4	GPIO 25	5 V Logic	Digital Out	Stepper Phase D
VCC (+)	VIN / 5V	5 V External	Power	External Motor Power Supply
GND (-)	GND	0 V	Ground	Common system ground

2. Component Setup & Details

2.1. SSD1306 OLED Display (128 × 64, I²C)

- **Connections:**
 - SDA → GPIO 21
 - SCL → GPIO 22
 - VCC → 3.3V
 - GND → GND
- **Notes:** The firmware configures the I²C bus clock to 400 kHz (`Wire.setClock(400000)`) on these pins to achieve a smooth 30 FPS frame rate for the character animation without lagging.

2.2. MFRC522 RFID Reader (SPI)

- **Connections:**
 - SDA (SS) → GPIO 5
 - SCK → GPIO 18
 - MISO → GPIO 19
 - MOSI → GPIO 23
 - RST → GPIO 4
 - 3.3V → 3.3V
 - GND → GND
- **Critical Warning:** The RC522 IC is strictly rated for 3.3 V. Connecting the power line to the 5 V or VIN terminal will permanently damage the RFID board.

2.3. HW-484 Hall Effect Sensor Module

- **Connections:**
 - DO (Digital Out) → GPIO 27
 - VCC → 3.3V
 - GND → GND
- **Operation:** Attached to a hardware interrupt (`hallISR`) on GPIO 27 . When a magnet on the tonearm approaches the sensor, DO drops to LOW , signaling that the tonearm is lowered.

2.4. ULN2003 Stepper Driver & 28BYJ-48 Motor

- **Connections:**

- IN1 → GPIO 26
- IN2 → GPIO 32
- IN3 → GPIO 33
- IN4 → GPIO 25
- VCC (+) → VIN / 5V (External 5 V Source)
- GND (-) → GND (Shared Ground)
- **Operation:** Uses an 8-step half-stepping sequence driven on Core 0. To prevent brownouts and signal noise, the motor power supply should be connected to VIN (5 V USB power) or a dedicated 5 V adapter.

3. Power Distribution & Safety Checklist

1. **Common Ground Line:** Ensure that all components share a single, unified GND path. A missing ground connection between the stepper motor driver and the ESP32 can cause floating signal voltages and erratic motor movement.
2. **Current Draw Consideration:** The 28BYJ-48 stepper motor draws up to 300 mA under load. Do not attempt to power the ULN2003 driver board from the ESP32's 3.3 V voltage regulator pin.
3. **Decoupling:** If experiencing power supply dips when the motor starts spinning, place a 100 μF electrolytic capacitor across the 5 V and GND terminals near the ULN2003 driver board.